

# Solid-State Relay

Lecture 17

# Electrical Relay

- A **relay** is a switch that allows a low-power signal to control a higher-power circuit.
- **Electromechanical relay (EMR):** Uses a coil and moving contacts.
- **Solid-state relay (SSR):** Uses semiconductor devices, no moving parts.

# Solid-State Relay

- A solid-state relay (SSR) is an electronic switching device that switches on or off when an external voltage (AC or DC) is applied across its control terminals.
- They serve the same function as an electromechanical relay, but solid-state relays contain no moving parts and have a longer operational lifetime.
- SSRs consist of a sensor which responds to an appropriate input (control signal), an electronic switching device that switches power to the load circuitry, and a coupling mechanism to enable the control signal to activate this switch without mechanical parts.



They may be designed to switch either AC or DC loads.

SSRs use power semiconductor devices such as thyristors and transistors , to switch currents up to around a hundred amperes.

SSRs have fast switching speeds compared with electromechanical relays, and have no physical contacts to wear out.

SSRs are unable to withstand a large momentary overload the way an electromechanical relay can, and have a higher "on" resistance.

Modern SSRs increasingly integrate built-in diagnostics and protection features, such as over-temperature shutoff, load monitoring, and short-circuit detection. These embedded protections help extend relay lifespan and prevent damage to connected loads or upstream circuitry, especially in industrial automation settings

# Operation

- In AC circuits, SCR or a triac relays switch off at the points of AC zero cross when there is zero load current.
- The circuit will never be interrupted in the middle of a sine wave peak, preventing the large transient voltages that would otherwise occur due to the sudden collapse of the magnetic field around the inductance. With the addition of a zero-point detector (and no adverse circuit inductance and resultant back EMF), the individual SCRs can be switched back on at the start of a new wave. This feature is called zero-crossing, or zero-crossover, switching.

An SSR based on a single MOSFET , or multiple MOSFETs in a paralleled array, can work well for DC loads.

MOSFETs have a characteristic of a reverse diode that conducts in the reverse direction, so a single MOSFET cannot block current in both directions.

For AC (bi-directional) operation, two MOSFETs are arranged back-to-back with their source pins tied together.

Multiple MOSFETs can be wired in parallel if switching a DC load. Usually a network is provided to speed the turn-off of the MOSFET when the control input is removed.

SSRs for DC switching applications may use MOSFETs or IGBTs.

Selection of the appropriate type of SSR is important, especially when the application calls for time critical On/Off condition with no variation.

Applications which require Time Critical On/Off switching, should use Transistor or MOSFET design types as they are not subject to the inherent zero-cross variations that SCR or TRIAC devices will exhibit.

The control signal must be coupled to the controlled circuit in a way that provides isolation between the two circuits.

Many SSRs use optical coupling. The control voltage energizes an internal LED , which illuminates and activates a photo-sensitive diode (photo-voltaic); the diode current turns on a back-to-back thyristor (TRIAC, SCR, or MOSFET) to switch the load. The optical coupling allows the control circuit to be electrically isolated from the load.



# Solid-State Relay

- A **solid-state relay** is an electronic switching device that:
- Activated by a low-voltage control signal.
- Switches a higher-voltage or higher-current load
- Uses **semiconductors instead of mechanical contacts**

# Basic Internal Structure of an SSR

- Most SSRs contain three main sections:
- **1. Input (Control Side)**
- Usually a **LED**
- Designed to work with logic signals (e.g., 3–32 V DC)
- When current flows through the LED, the relay turns on.

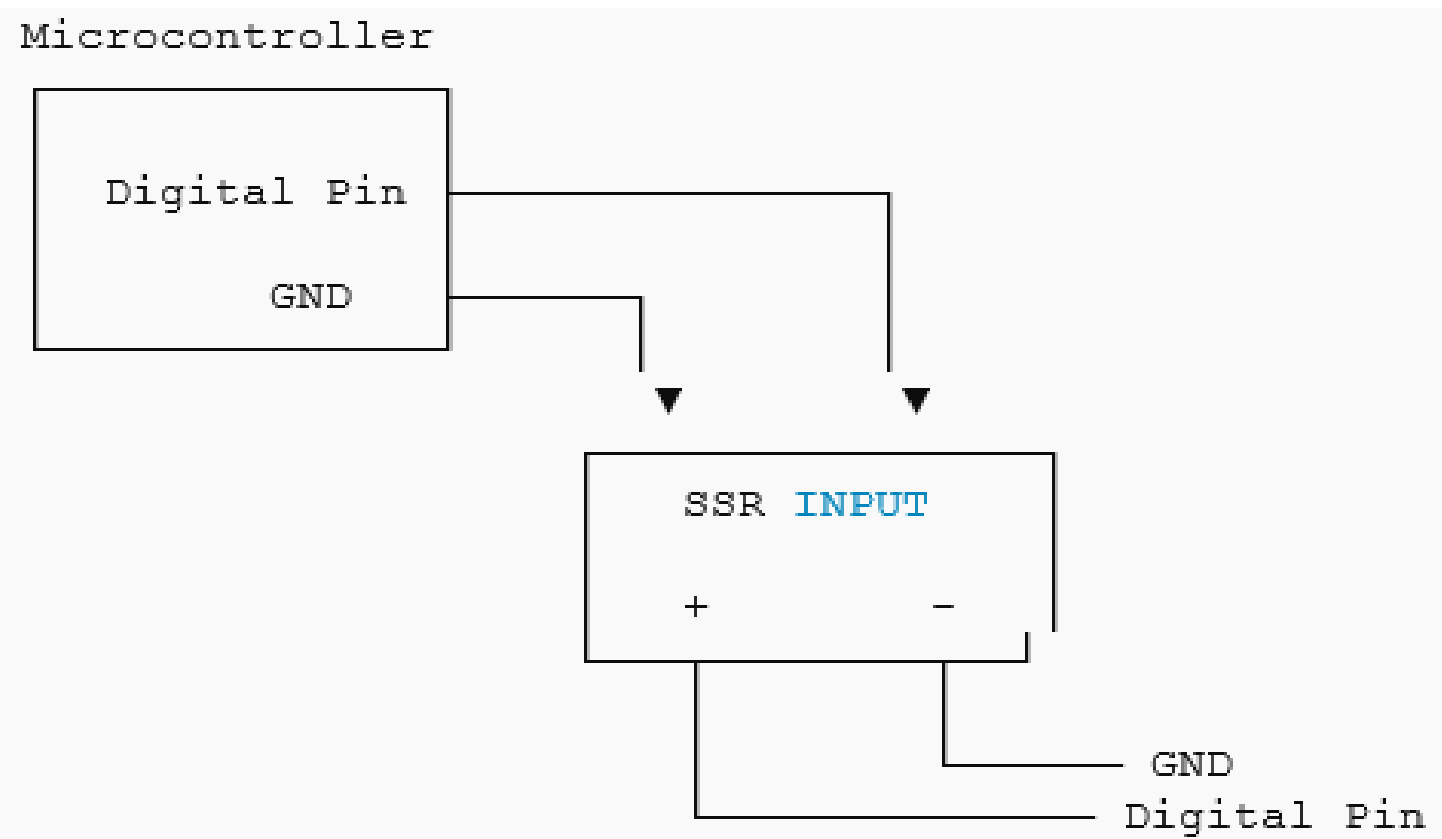
## 2. Isolation (Optocoupler)

- Light from the LED activates a photosensitive device.
- Provides **electrical isolation** between control and load.
- Protects low-voltage electronics from high voltage.

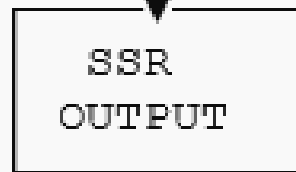
## 3. Output (Switching Side)

Depends on the type of SSR:

- **Triac / SCR** → AC loads
- **MOSFET / Transistor** → DC loads



AC MAINS  
(HOT / LINE)



(Lamp, Heater, Motor\*)



AC NEUTRAL

# Types of Solid-State Relays

- **A. AC Output SSRs**

Use **triacs or SCRs**

Common for:

- Heaters
- Lamps
- AC motors

Often includes **zero-cross switching**

- **Zero-cross switching:**

Turns on only when AC voltage crosses zero.

Reduces electrical noise.

## **B. DC Output SSRs**

- . Use **MOSFETs or BJTs**
- . Used for:
  - . DC motors
  - . Solenoids
  - . Battery-powered systems
- . Faster and more precise than AC SSRs.

## **C. Random-Turn-On SSRs**

Turn on immediately when the control signal is applied.

Used in:

Phase control

Light dimming

Motor speed control

# **Advantages of Solid-State Relays**

## **Reliability**

- No moving parts → no mechanical wear.
- Very long operational life.

## **Silent Operation**

- No clicking sounds.

## **Fast Switching**

- Microseconds instead of milliseconds.
- Ideal for high-speed control.

## **Electrical Isolation**

- Protects sensitive electronics.

## **Resistant to Shock and Vibration**

- Useful in industrial or mobile environments.



# Disadvantages of Solid-State Relays

- **Heat Generation**

Voltage drop across semiconductors.

Often require **heat sinks**.

- **Leakage Current**

Even when “off,” a small current flows.

Can cause a faint glow in LEDs or neon lamps.

- **Cost**

Usually more expensive than EMRs.

- **Failure Mode**

Often fails short (**ON**) instead of open.

Can be a safety concern.

# Comparison between SSR and Electromechanical Relay

| Feature         | <u>SSR</u> | <u>EMR</u> |
|-----------------|------------|------------|
| Moving parts    | No         | Yes        |
| Speed           | Very fast  | Slower     |
| Noise           | Silent     | Clicks     |
| Lifetime        | Very long  | Limited    |
| Leakage current | Yes        | No         |
| Heat            | Higher     | Lower      |

# Important Ratings of SSR

- When selecting an SSR, the following ratings should be considered:

- ◆ **Control Voltage**

Example: 3–32 V DC

- ◆ **Load Voltage**

Example: 24–480 V AC or 0–60 V DC

- ◆ **Load Current**

- Maximum current the SSR can handle.

- ◆ **Turn-On Method**

Zero-cross or random

- ◆ **Thermal Requirements**

- A heat sink is needed for certain currents.

# Typical Applications for SSR

- Industrial automation.
- Temperature controllers.
- Motor control.
- Lighting systems.
- HVAC systems.
- Robotics and microcontroller projects.

# Choose the Correct SSR

- The SSR should match the load specifications.

## **A. Input (Control Side):**

- Input voltage range (e.g. **3–32 V DC**)
- Input current (often 5–20 mA)
- Make sure that the microcontroller pin can supply enough current.

**B. Output (Load Side):**

Based on the load specifications:

| Load Type           | SSR Type        |
|---------------------|-----------------|
| AC lamp / heater    | AC SSR (triac)  |
| DC motor / solenoid | DC SSR (MOSFET) |
| Precise timing      | Random-turn-on  |
| Reduced noise       | Zero-cross      |

# Zero-Cross SSR Timing Considerations

For a **zero-cross SSR**:

- Turning ON happens at the **next AC zero crossing**.
- Turning OFF happens when the current drops below the holding current.

# Power and Heat Considerations

- SSRs **dissipate heat** even when fully on.
- **Example**
- $1.6 \text{ V drop} \times 10 \text{ A} = \mathbf{16 \text{ W of heat}}$



# Comparison of Solid-State Relays (SSRs) with Contactors

A **contactor** is a **heavy-duty electromechanical relay** designed to:

- Switch **high currents and voltages**.
- Control motors, heaters, and power distribution.
- Operate safely under frequent load switching.
- It uses:

An **electromagnetic coil**.

**Mechanical contacts** that open and close physically.

# 1. Core Operating Difference

| Aspect           | SSR                   | Contactor           |
|------------------|-----------------------|---------------------|
| Switching method | Semiconductor devices | Mechanical contacts |
| Moving parts     | None                  | Yes                 |
| Isolation        | Optical               | Physical air gap    |
| Switching sound  | Silent                | Audible “clunk”     |

# 2. Electrical Performance Comparison

## A. Current & Voltage Capability

| Feature                | SSR         | Contactor                        |
|------------------------|-------------|----------------------------------|
| Typical current range  | mA → ~100 A | 10 A → 1000+ A                   |
| Surge tolerance        | Limited     | Very high                        |
| Short-circuit handling | Poor        | Excellent (with arc suppression) |

## B. On-State Losses

| Feature            | SSR              | Contactor |
|--------------------|------------------|-----------|
| On-state drop      | 1–2 V (AC triac) | ~0 V      |
| Heat generation    | High             | Very low  |
| Heat sink required | Often            | Never     |

SSRs waste power as heat; contactors don't.

# C. Switching Speed

| Feature                | SSR               | Contactor |
|------------------------|-------------------|-----------|
| Turn-on time           | $\mu\text{s}$ –ms | 10–50 ms  |
| Switching frequency    | Very high         | Low       |
| Best for rapid cycling | Yes               | No        |

### 3. Load Type Compatibility

| SSR                                                | Contactor                                            |
|----------------------------------------------------|------------------------------------------------------|
| Resistive loads (heaters)                          | Motors                                               |
| Light inductive loads                              | Transformers                                         |
| High-cycle switching                               | Capacitive and inductive loads                       |
| Sensitive to Large motors and High inrush currents | Designed to handle:<br>Inrush currents<br>Arc energy |

# 4. Lifetime & Reliability

| Feature          | SSR                      | Contactor         |
|------------------|--------------------------|-------------------|
| Mechanical wear  | None                     | Yes               |
| Electrical wear  | Junction aging           | Contact erosion   |
| Typical lifetime | Millions–billions cycles | 100 k–10 M cycles |

# 5. Failure Modes (Very Important)

| Failure Mode        | SSR              | Contactor |
|---------------------|------------------|-----------|
| Fail open           | Rare             | Common    |
| Fail closed         | Common           | Rare      |
| Visible state       | No               | Yes       |
| Safety implications | Can be dangerous | Safer     |



## 6. Control Requirements

| Feature          | SSR               | Contactor      |
|------------------|-------------------|----------------|
| Control voltage  | 3–32 V DC typical | 12–240 V AC/DC |
| Control current  | Few mA            | Hundreds of mA |
| MCU direct drive | Yes               | Usually no     |

## 7. Noise

| Feature          | SSR              | Contactor     |
|------------------|------------------|---------------|
| Audible noise    | None             | Loud          |
| Electrical noise | Low (zero-cross) | High (arcing) |

## 8. Cost & Size

| Feature              | SSR                | Contactor         |
|----------------------|--------------------|-------------------|
| Initial cost         | Higher (low power) | Lower (low power) |
| Cost at high current | Very high          | Reasonable        |
| Physical size        | Compact            | Large             |

# Summary Table

| Category         | SSR                | Contactor       |
|------------------|--------------------|-----------------|
| Speed            | Very fast          | Slow            |
| Noise            | Silent             | Loud            |
| Power handling   | Limited            | Very high       |
| Heat             | High               | Low             |
| Safety isolation | Electronic         | Physical        |
| Best for         | High-cycle control | Power switching |